

An eight-task pilot of task-level predictability for heterogeneous decision primitives

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Abstract

When a system must make a bounded decision, it can implement that decision as a rule, a conventional classifier, a typed decision model, or a generative language model. Those primitives differ in quality and in operational behavior. A natural hope is that measurable properties of the task, fixed before any primitive is scored, would predict that behavior on a task the predictor has not seen.

We report an eight-task pilot that freezes the task set, the fingerprint, the primitive implementations, and the predictor before the predictive results are known. The target is a seven-coordinate profile: macro-F1, local training seconds, local inference seconds, API dollars, per-example latency at the 50th and 95th percentiles, and the invalid-output rate. The predictor is seven independent ridge regressions with penalty 1. The baseline is the mean profile of the same primitive on the tasks used to fit that fold. ESCI is held out of every fit as an e-commerce domain. Other holdouts remove a task family. Several of those families contain only one task.

On the measured matrix, quality and validity vary sharply by primitive and task. Under the pinned Qwen2.5-0.5B configuration, the invalid-output rate runs from 0.023 and 0.025 on VitaminC and SMS Spam to 1 on Civil Comments, where macro-F1 is 0 because every output failed the one-label rule. The ridge model is closer than the primitive mean on macro-F1 for 5 of 8 held-out tasks and on the invalid rate for 2 of 8. Under every vector policy, both predictors select the classifier. On ESCI, macro-F1 error falls from 0.2825 to 0.1024 and invalid-rate error from 0.1199 to 0.0444, and the selected primitive does not change.

Eight tasks do not establish that task fingerprints are unpredictable in general, that one primitive is best, or that latency on this rig is a hardware-normalized speed. The rule, the classifier, and the decision model were timed on CPU. The generative model was timed on a GTX 1650 Ti in float16. API dollars are 0 because no API was called. Semantic Scholar returned HTTP 429 during the literature screen, and Scopus was not queried.

Keywords: bounded decisions, task fingerprint, computational primitives, measurement pilot.

1 Introduction

Bounded decisions are classifications with a declared label set: an intent, a toxicity bit, a spam bit, a clause type, a yes/no/maybe answer, a fact-checking label, or a product-relevance label. The same decision can be implemented in more than one way. A rule matches a pattern. A classifier is fit on labeled text. A typed decision model returns a distribution over a label set supplied at inference time. A generative model writes text that a parser then accepts or rejects as a label.

Algorithm selection already treats the choice of procedure as a map from problem features to performance (Rice, 1976; Smith-Miles, 2009). LLM routing already chooses among language models, often with cost in view, and sometimes with a notion of task type (Shnitzer et al., 2023; Ong et al., 2024; Liu et al., 2026a; Li, Zhang, et al., 2026b). Comparisons of encoders or linear classifiers with prompted generative models already report quality together with latency or dollars on fixed-label text tasks (Zhang et al., 2025; Gonzalez, 2026; Bucher and Martini, 2024; Vajjala and Shimangaud, 2025). Leave-one-dataset-out selection from training-split meta-features has been studied for abstractive summarization (Bino and Shrivastava, 2026). This pilot does not claim those problems as new.

What we measure is narrower. Each task is scored, under one input string and one label set, by the primitives that the protocol marks as eligible. A fingerprint is computed from the task contract and the training split only. A ridge model, specified before these predictive results, maps that fingerprint and a primitive identity to the measured profile. We then ask whether the predicted profile is closer to the measured profile than the training-fold mean of that primitive, and whether any closer prediction changes the primitive selected by a policy that was also specified in advance.

The independent unit for that second question is the task. The evaluation rows inside a dataset estimate that dataset’s scores. They do not multiply the number of tasks. With eight tasks, and with three of the five selector-training families containing a single task, the pilot cannot support a general claim about family-level transfer. We state that limit in the body of the paper.

The empirical reading is as follows. Computational primitives exhibit substantial task-dependent differences in quality and operational behavior, but in this eight-task pilot, a simple pre-deployment task fingerprint did not reliably improve prediction over primitive-specific training-fold mean baselines or alter the resulting computational choice. This does not say that the hypothesis is false, that the classifier is universally best, or that we have introduced a router.

2 Research questions

- RQ1. How heterogeneous are quality, validity, latency, and cost across computational primitives on these bounded decision tasks?
- RQ2. How much predictive information about primitive behavior, in this pilot, is contained in the fixed pre-deployment fingerprint?
- RQ3. Does that fingerprint, through the pre-specified ridge model, improve held-out prediction over the primitive-specific mean?
- RQ4. Does a closer predicted profile change the primitive selected by the pre-specified policies?
- RQ5. What failure modes appear when the pinned generative configuration is used as a bounded decision primitive?

RQ2 and RQ3 are about this fingerprint and this model. They are not questions about every possible fingerprint. RQ1 and RQ5 are descriptive. RQ4 separates prediction error from the downstream choice.

3 Related work

We organize prior work by the claim it already occupies. We do not treat the search as an exhaustive systematic review. On 26 September 2026, Semantic Scholar’s API returned HTTP 429, and Scopus was not queried. OpenAlex full-text search over 2024 onward was noisy. The papers below are

ones we retrieved and read far enough to place. Two further retrieved papers, on guarded query routing, on a preference platform for LLM routers, and on latent-task prompt routing, are listed in the bibliography and are not part of the comparison in this section (Šléher et al., 2025; Lai et al., 2026; Zhang et al., 2026).

Algorithm selection. Rice (1976) formulates selection as a map from problem features to algorithm performance. Smith-Miles (2009) surveys that map as meta-learning across disciplines. Any paper that says task features predict which procedure will perform well is inside this frame. Our primitive set is not that survey’s portfolio. The map itself is not a contribution of this pilot.

Routing among language models. Shnitzer et al. (2023) train routers on benchmark datasets and apply them to a new task. The router uses embeddings of the new task’s inputs. GraphRouter (Feng et al., 2024) represents tasks, queries, and LLMs in one graph, predicts performance and cost, and then applies a user preference. TRouter (Liu et al., 2026a) uses synthesized task profiles for cold-start routing among LLMs. LLMRouterBench (Li et al., 2026b) re-evaluates a large set of LLM routers and reports that many of them do not reliably beat a simple baseline, with a remaining gap to an oracle. RouteLLM, Hybrid LLM, RouterBench, and FrugalGPT select among generative models at query level (Ong et al., 2024; Ding et al., 2024; Hu et al., 2024; Chen et al., 2023). Those systems are neighbors. Their action is which language model to call. This pilot’s action includes procedures that do not generate.

Encoder or classifier versus a generative classifier. TaMAS argues from a manual task taxonomy that BERT-like models and LLMs win on different kinds of text classification, and it does not train a held-out selector (Zhang et al., 2025). Other studies compare fine-tuned encoders, linear models, or logistic regression with zero-shot or few-shot generative models and, in some cases, report latency and dollar cost (Bucher and Martini, 2024; Edwards and Camacho-Collados, 2024; Vajjala and Shimangaud, 2025; Gonzalez, 2026). Cascades that send uncertain classifier outputs to an LLM use a signal that exists only after a model has run (de Andrade et al., 2024; Abedini and Heydarnoori, 2025; Kondadadi and Ortega, 2026). Repeating the finding that a trained classifier can beat a prompted LLM on a fixed label set would not be a new result. We cite those papers and do not rediscover them.

Dataset-level meta-learning. Bino and Shrivastava (2026) extract meta-features from the training split of summarization datasets and use leave-one-dataset-out to predict which pretrained summarizer wins under a stated metric. That is the closest protocol we retrieved: features available without the test set, and the dataset as the holdout unit. Their candidates are abstractive summarizers. They predict a winner under one metric. They do not predict a vector that includes an invalid-output rate, and they do not include a rule, a conventional text classifier, a typed decision model, and a generative decision model together.

Typed decision models and output constraints. Laya is an open decision-model checkpoint (convaiinnovations, 2026). Public descriptions of Jev and Decision 1.0 treat typed decisions as a software interface (Liu et al., 2026b; Li et al., 2026a). We use Laya as one pinned implementation. We do not treat vendor numbers as results. Separately, constrained decoding can remove structural schema failures of small generative models (Chavan, 2026). Our generative pin does not use that constraint. The invalid rates below are rates under exact string match, not rates under grammar-constrained decoding.

No paper in this retrieved set trains on a fixed pre-deployment fingerprint and predicts the seven-coordinate profile of a rule, a TF-IDF logistic regression, Laya, and Qwen2.5-0.5B-Instruct under family holdouts and an e-commerce holdout. That is a statement about the sources we obtained. It is not a claim that the experiment is absent from every index, and it is not a claim of priority.

4 Problem formulation

For a task t and an eligible primitive p , the measured profile is

$$Y_{t,p} = (Q, C_{\text{train}}, C_{\text{infer}}, C_{\text{api}}, L_{p50}, L_{p95}, V). \quad (1)$$

Q is macro-F1 on the frozen evaluation rows. Accuracy is stored and is not the quality coordinate. C_{train} and C_{infer} are local training and inference seconds. C_{api} is API dollars. L_{p50} and L_{p95} are per-example latency in milliseconds, excluding model load. V is the fraction of evaluation rows whose output is not exactly one legal label.

An invalid row is already incorrect for macro-F1. V is kept as its own coordinate because a legal wrong label and an output that is not a label are different failures, and macro-F1 does not say which occurred.

The predictor’s job is

$$(\text{fingerprint of } t, \text{ identity of } p) \mapsto \hat{Y}_{t,p}. \quad (2)$$

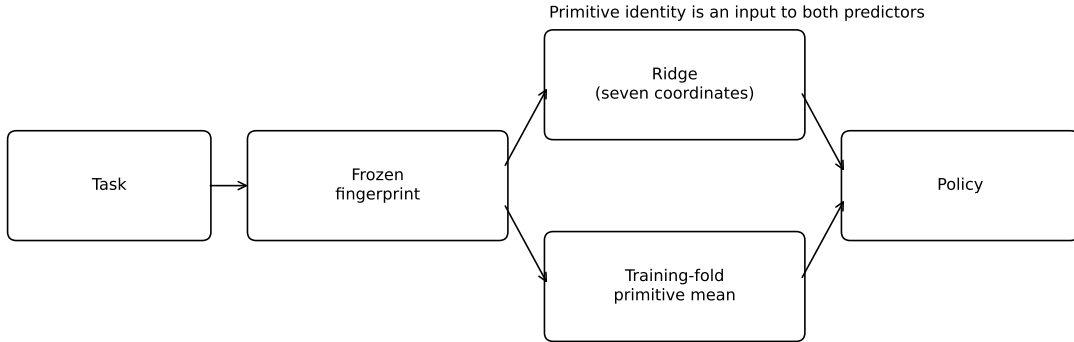


Figure 1: A task yields the frozen fingerprint. Primitive identity enters both the seven ridge models and the training-fold primitive mean. A policy reads a predicted profile. There is no arrow from the task directly to a winning primitive.

The fingerprint is shared across primitives. The primitive identity is an intercept shift, with the classifier as the reference level. There is no product between a fingerprint coordinate and a primitive indicator. A policy may read $\hat{Y}$ after it is predicted. The policy is not the training label. This split is deliberate: one utility would bake a preference into the fit, and encoder-versus-LLM work already varies such weights (Gonzalez, 2026).

Higher Q is better. Lower cost, latency, and V are better. Ineligible pairs have no target. The rule is eligible only on SMS Spam. A measured zero stays zero. Training seconds for the rule, Laya,

and Qwen are 0 because those primitives are not fit on the task. API dollars are 0 because no API was called.

5 Computational primitives

All eligible primitives on a task see the same input string and the same label set.

Rule. A single regular expression, written and hashed before scores, maps an SMS message to spam on a match and to ham otherwise. SHA-256 of the file bytes, including the trailing newline, is

264e288b89a430005de7b72fdfeb77862e46b23b7415c76df285e4e5e1a7f992.

The expression is not fit on labels. No other task receives a rule. A rule is not added to a 77-, 100-, or 151-way label set because another primitive was weak.

Classifier. Word unigrams and bigrams, `min_df=2`, at most 30,000 features, sublinear term frequency, and logistic regression with the saga solver, $C = 1$, and at most 200 iterations. The iteration limit was not raised after a convergence warning on SMS Spam. Tasks with at most 100,000 fingerprint training rows use every such row (Cfull). Civil Comments, VitaminC, and ESCI are fit on a stratified 100,000-row subset (C100k). That cap is a pre-registered compute limit. It is not a scarcity experiment and it is not the full-data classifier.

Typed decision model. `convaiinnovations/laya`, English router, one choice question. Criteria keys are the sorted training labels. Descriptions equal those keys. The instruction is “Which label applies?” Context limits are `max_len=2048` and `head_max_len=1024`. The run is on CPU. Load time is stored outside latency.

Generative model. `Qwen/Qwen2.5-0.5B-Instruct`, `do_sample` false, at most 32 new tokens, float16 on a GTX 1650 Ti. The prompt asks for exactly one label from a sorted list and forbids an explanation. An output is valid only if, after stripping whitespace and at most one matching pair of surrounding quotes, it equals a legal label. Anything else is invalid. This configuration was not retuned after the invalid rates were observed.

6 Task fingerprint

The fingerprint uses the task contract and the training split only. It does not use test labels, test examples, any primitive’s outputs, or any coordinate of Y . It does not use the benchmark name, the dataset identifier, or a hand rating of difficulty.

Token length is `bert-base-uncased` WordPiece without special tokens. The revision is

86b5e0934494bd15c9632b12f734a8a67f723594.

Label-name similarity is mean-pooled `all-MiniLM-L6-v2`. The revision is

1110a243fdf4706b3f48f1d95db1a4f5529b4d41,

after lowercasing and replacing underscores and hyphens with spaces. Normalized entropy is Shannon entropy in nats divided by $\ln K$.

The ridge model receives, in this order: $\log_{10}$ of the training-row count, $\log_{10}$ of K , normalized entropy, $\log_{10}$ of the max/min class-count ratio, $\log_{10}$ of the minimum and of the median rows per class, $\log_{10}$ of the median and P95 lengths, the mean and minimum label-name cosines, and a bit that is 1 when the relation code is `paired_text`. Input structure and evidence required make the same split of these eight tasks as that bit, so they are not entered twice. External knowledge is false on every task. Documented shift is `unspecified` on four selector-training tasks and blank on VitaminC, so it is recorded and not used as a feature. The rules flag is true only on SMS Spam and is not a feature. The primitive mask decides eligibility instead.

Civil Comments and ESCI lengths are medians and P95s on 20,000 training rows drawn with seed 20260926. Their imbalance uses the full training split. Tables 1a and 1b report the fingerprint values used by the study.

7 Experimental protocol

Evaluation rows are drawn before any fit. If the official test side has at most 1,000 rows, all of them are used. Otherwise 1,000 rows are drawn without replacement. PubMedQA’s test half has 500 rows, so its evaluation count is 500. Every other task in Table 2 has 1,000 evaluation rows.

The ridge specification was frozen before this fit. Each coordinate has its own `Ridge(alpha=1)` model with an intercept. Numeric fingerprint columns are standardized with the population mean and standard deviation of the training tasks in that fold. A column with zero variance is set to 0, including on the held-out task. Primitive indicators are not standardized. Predictions of Q and V are clipped to $[0, 1]$. Time coordinates are clipped at 0. The penalty was not searched. No interaction was added after the errors were seen.

API dollars are predicted as 0. Training seconds of the rule, Laya, and Qwen are predicted as 0 from the primitive definition. The training-seconds regression is fit only on classifier rows.

The baseline ignores the fingerprint. For each primitive present in the training tasks, it predicts the arithmetic mean of that primitive’s measured coordinate. An unseen primitive is not predicted and is not imputed from another primitive. On the security holdout, SMS Spam is removed, so the rule slot is unpredictable.

Task-level error is the mean absolute error across eligible primitives that both predictors can fill. Coordinates are not averaged into one score. A negative difference means the ridge prediction was closer.

Policies run after a vector exists. Always-Qwen, always-classifier, always-Laya, and the mask default do not read a vector. The mask default selects the rule when it is eligible and the classifier otherwise. The other policies read a predicted vector, or the measured vector as an oracle that is not a deployable predictor: highest macro-F1; lowest L_{p50} ; equal weights $U = \tilde{Q} - \tilde{C}_{\text{infer}} - \tilde{L}_{p50}$; shared weights $U = \tilde{Q} - \frac{1}{2}\tilde{C}_{\text{infer}} - \frac{1}{2}\tilde{L}_{p50}$. Tildes are min-max maps whose minimum and maximum come from the training tasks of that fit, not from the held-out task. Ties break in the order rule, classifier, decision model, generative model. V is reported with the chosen primitive and is not given a weight.

Family labels are the labels already written in the screening record. They are not features. Intent holds out Banking77 and CLINC150 plus together. Verification holds out PubMedQA and VitaminC together. Moderation, security, and business/legal each hold out one task. The e-commerce holdout fits on all seven selector-training tasks and scores ESCI. ESCI does not enter any fit, any primitive mean, any standardization, or any utility range. A random 70/15/15 split of tasks is not run. Seven

tasks do not fill those fractions.

Example-level uncertainty is the uncertainty of a score on one dataset’s evaluation rows. Task-level uncertainty is the uncertainty of a comparison across tasks. Family-level uncertainty would require several tasks in the held-out family and several tasks left behind that still represent the fingerprint’s informative bits. This design has eight tasks. Three family holdouts are single-task holdouts. The verification holdout removes both `paired_text` training tasks, so that bit has no variance and is zeroed. We do not report p -values. A count such as “5 of 8” is a description of this sample, not a population rate.

8 Datasets and task families

Tables 1a and 1b are the frozen fingerprint, split into two parts so the type stays readable. Selector training is yes except for ESCI. Relation `paired_text` marks tasks whose input joins two texts. The others are single utterances. The releases are those named in the screening record (Casanueva et al., 2020; Larson et al., 2019; Civil Comments; SMS Spam Collection; Chalkidis et al., 2021; Jin et al., 2019; Schuster et al., 2021; Reddy et al., 2022).

Table 1a: Training size, imbalance, and relation. Lengths are in Table 1b. Imbalance uses the full training split. Entropy is normalized entropy. The source file is named in Section 14.

Task	Family	Train rows	K	Entropy	Max/min	Relation	Selector
Banking77	Intent	10003	77	0.9920	5.343	utterance	yes
CLINC150 plus	Intent	15250	151	0.9990	2.500	utterance	yes
Civil Comments	Moderation	1804874	2	0.4021	11.505	utterance	yes
SMS Spam	Security	4458	2	0.5681	6.467	utterance	yes
LEDGAR	Business / legal	60000	100	0.9348	137.696	utterance	yes
PubMedQA fold 0	Verification	450	3	0.8516	5.082	paired text	yes
VitaminC	Verification	370653	3	0.9030	3.505	paired text	yes
ESCI	E-commerce	1393063	4	0.6228	33.310	paired text	no

Table 1b: Input length and label-name similarity. Lengths for Civil Comments and ESCI are on the pre-registered 20,000-row sample. Entropy in Table 1a is normalized entropy.

Task	Median tokens	P95 tokens	Mean cosine	Min cosine
Banking77	11.0	35.0	0.2011	−0.1309
CLINC150 plus	9.0	15.0	0.1342	−0.1968
Civil Comments	46.0	201.0	0.5701	0.5701
SMS Spam	18.0	53.0	0.2830	0.2830
LEDGAR	104.0	379.0	0.2194	−0.1086
PubMedQA fold 0	319.5	490.1	0.5272	0.4000
VitaminC	45.0	89.0	0.2113	0.1966
ESCI	213.0	805.0	0.2643	0.1075

Civil Comments is binarized at toxicity 0.5. The input is comment text only. PubMedQA training rows are fold 0 of the official seed-0 split, 450 rows, not the undivided 1,000-row pool. ESCI rows are English (US), large version, from the official examples file. Product text for the length sample and for model inputs was read from a tasksource copy keyed by product id and locale, because the official products file was not fully downloaded. That provenance is part of the

measurement, not a silent substitution of the row set. The official examples filter matches 1,393,063 training rows.

9 Results

9.1 Heterogeneity

Table 2 is the performance matrix. Blank primitive-task pairs were not eligible. Nothing in the table was filled with zero in place of a missing run. Cfull and C100k are the classifier. Device is part of the record.

Table 2: Measured profiles. Q is macro-F1. Train and infer are seconds. API is dollars and is 0 because no API was called. Latency is milliseconds and does not include model load. Invalid is a count. The rule, the classifier, and Laya are CPU. Qwen is a GTX 1650 Ti in float16. C100k marks the pre-registered 100,000-row classifier fit. Source: [research/docs/performance_matrix.md](#).

Task	Primitive	Q	Time (s)		API (\$)	Latency (ms)		Invalid	n	Device
			Train	Infer		p50	p95			
SMS Spam	Rule	0.8576	0.000	0.028	0	0.020	0.062	0	1000	CPU
SMS Spam	Classifier	0.9393	0.295	0.351	0	0.188	0.719	0	1000	CPU
SMS Spam	Laya	0.7822	0.000	724.349	0	698.724	941.798	0	1000	CPU
SMS Spam	Qwen	0.2067	0.000	308.276	0	323.515	417.066	25	1000	GPU
PubMedQA	Classifier	0.2691	0.635	0.194	0	0.261	0.623	0	500	CPU
PubMedQA	Laya	0.3128	0.000	1019.183	0	1978.849	2842.361	0	500	CPU
PubMedQA	Qwen	0.2327	0.000	474.746	0	932.252	1239.578	174	500	GPU
Banking77	Classifier	0.8508	2.412	0.267	0	0.184	0.456	0	1000	CPU
Banking77	Laya	0.3888	0.000	5260.039	0	5212.193	5888.881	0	1000	CPU
Banking77	Qwen	0.1316	0.000	1360.532	0	1274.864	1668.698	514	1000	GPU
CLINC150 plus	Classifier	0.8299	5.033	0.237	0	0.177	0.337	0	1000	CPU
CLINC150 plus	Laya	0.4280	0.000	4614.129	0	4584.007	4905.017	0	1000	CPU
CLINC150 plus	Qwen	0.2465	0.000	1455.811	0	1336.641	1980.492	745	1000	GPU
LEDGAR	Classifier	0.7418	139.867	0.298	0	0.215	0.481	0	1000	CPU
LEDGAR	Laya	0.2088	0.000	4724.659	0	4528.972	6190.025	0	1000	CPU
LEDGAR	Qwen	0.0281	0.000	1346.729	0	1236.580	1887.759	86	1000	GPU
Civil Comments	Classifier (C100k)	0.6298	19.757	0.316	0	0.196	0.628	0	1000	CPU
Civil Comments	Laya	0.7367	0.000	807.656	0	726.485	1376.470	0	1000	CPU
Civil Comments	Qwen	0.0000	0.000	544.926	0	489.657	808.857	1000	1000	GPU
VitaminC	Classifier (C100k)	0.3826	15.300	0.251	0	0.176	0.416	0	1000	CPU
VitaminC	Laya	0.2165	0.000	797.699	0	766.750	1045.356	0	1000	CPU
VitaminC	Qwen	0.2209	0.000	603.543	0	578.279	779.076	23	1000	GPU
ESCI	Classifier (C100k)	0.2189	77.896	0.253	0	0.181	0.394	0	1000	CPU
ESCI	Laya	0.1527	0.000	1909.519	0	1444.679	4742.117	0	1000	CPU
ESCI	Qwen	0.0358	0.000	1024.658	0	922.247	2060.757	32	1000	GPU

Table 3: Qwen invalid-output rate. Rate is the invalid count divided by the evaluation count n . The rule, the classifier, and Laya have zero invalid outputs on every scored task.

Task	Invalid / n	Rate
VitaminC	23 / 1000	0.023
SMS Spam	25 / 1000	0.025
ESCI	32 / 1000	0.032
LEDGAR	86 / 1000	0.086
PubMedQA	174 / 500	0.348
Banking77	514 / 1000	0.514
CLINC150 plus	745 / 1000	0.745
Civil Comments	1000 / 1000	1.000

Macro-F1 on SMS Spam runs from 0.2067 (Qwen) to 0.9393 (classifier), with the rule at 0.8576 and Laya at 0.7822. On Banking77 the classifier reaches 0.8508 and Qwen 0.1316. On LEDGAR the classifier reaches 0.7418 and Qwen 0.0281. The ordering is not the same on every task. On PubMedQA, Laya’s macro-F1 is 0.3128 and the classifier’s is 0.2691. On Civil Comments, Laya’s is 0.7367 and the C100k classifier’s is 0.6298. On VitaminC, the classifier (0.3826) is ahead of Qwen (0.2209) and Laya (0.2165), and Qwen’s invalid rate is only 0.023, so that gap is mostly legal predictions. We do not collapse these rows into a ranking.

Local inference time and latency are not comparable across devices. Classifier p50 latency is under 0.3ms on every task. Laya’s p50 on CPU runs from 698.7ms on SMS Spam to 5212ms on Banking77. Qwen’s p50 on the GTX 1650 Ti runs from 323.5ms on SMS Spam to 1336.6ms on CLINC150 plus. Those numbers characterize the deployments we ran. They do not establish which primitive is intrinsically faster. API dollars do not vary.

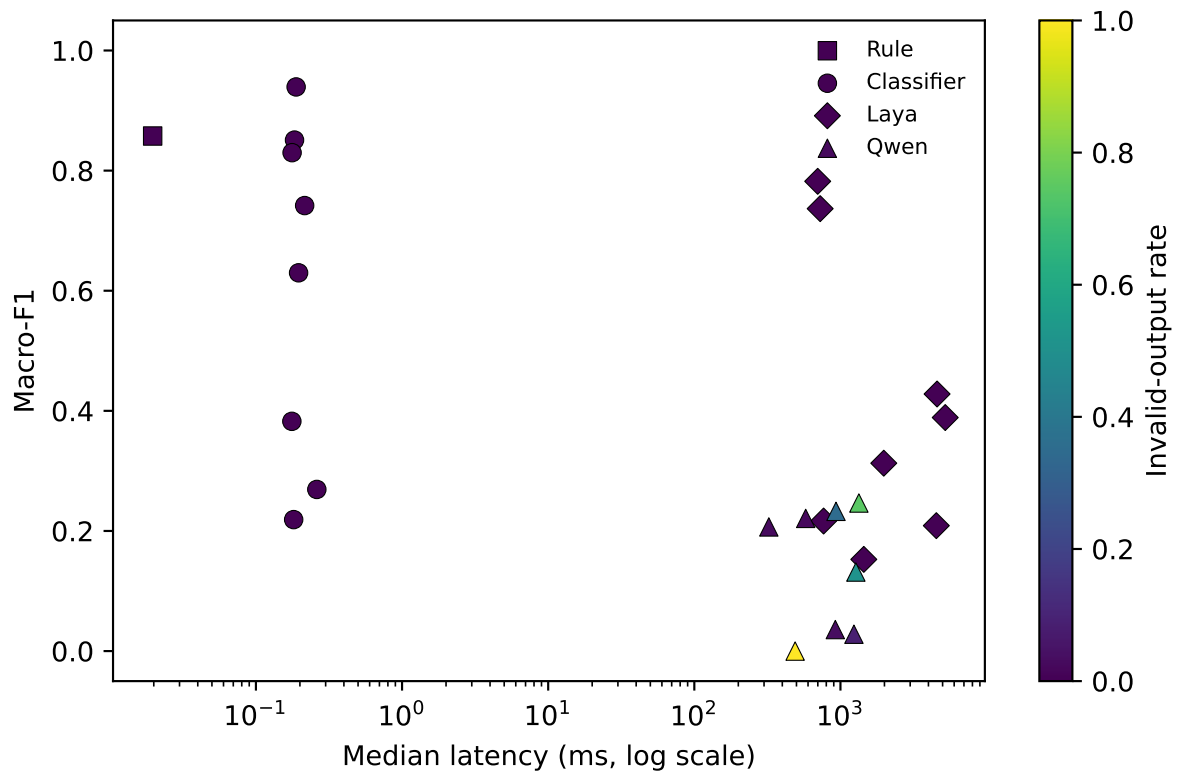


Figure 2: Measured macro-F1 against median latency on a log scale, one marker per eligible pair in Table 2. Marker shape is the primitive. Color is the invalid-output rate. The rule, the classifier, and Laya are CPU timings. Qwen is the GTX 1650 Ti in float16. There is no trend line and no confidence band.

Training seconds are nonzero only for the classifier, from 0.295 s on SMS Spam to 139.867 s on LEDGAR. The C100k fits are 19.757 s (Civil Comments), 15.300 s (VitaminC), and 77.896 s (ESCI).

9.2 Prediction against the primitive mean

Table 4 is the task-level mean absolute error. Each cell averages the eligible primitives both predictors could fill. For the security holdout that set excludes the rule.

Table 4: Held-out absolute error, ridge versus the training-fold primitive mean. Each cell averages the eligible primitives both predictors could fill. For the security holdout that set excludes the rule. Source: `research/docs/selector_results.md`.

Holdout	Task	Q ridge	Q mean	V ridge	V mean
Intent	Banking77	0.2112	0.1090	0.1713	0.0725
Intent	CLINC150 plus	0.1623	0.1232	0.2483	0.1495
Verification	PubMedQA	0.2583	0.2785	0.0792	0.0420
Verification	VitaminC	0.2096	0.2688	0.2839	0.1503
Moderation	Civil Comments	0.2075	0.1880	0.3333	0.2366
Security	SMS Spam	0.2271	0.2619	0.2832	0.1426
Business / legal	LEDGAR	0.1562	0.1684	0.1131	0.1188
E-commerce	ESCI	0.1024	0.2825	0.0444	0.1199

Ridge is closer on macro-F1 for PubMedQA, VitaminC, SMS Spam, LEDGAR, and ESCI. It is farther on Banking77, CLINC150 plus, and Civil Comments. Ridge is closer on the invalid rate only for LEDGAR and ESCI.

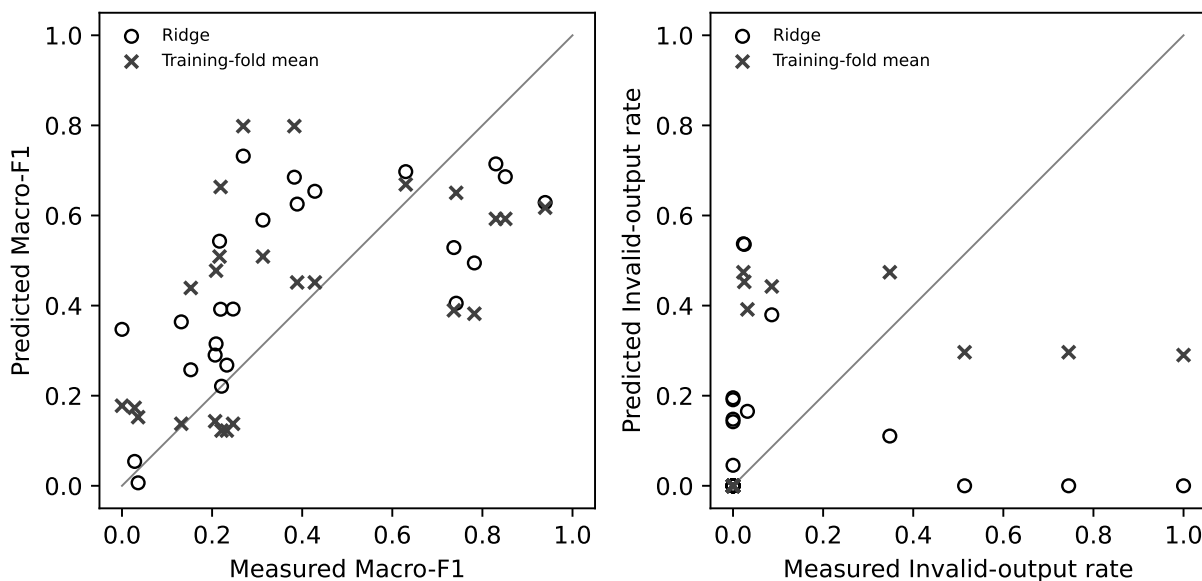


Figure 3: Predicted versus measured macro-F1, and predicted versus measured invalid rate, for held-out pairs both predictors filled. The diagonal is equality. Open circles are ridge. Crosses are the training-fold mean.

Median-latency error is smaller for ridge on four tasks and smaller for the mean on four. Those errors are hundreds of milliseconds. They are reported in the result file and are not a hardware-normalized comparison. API-dollar error is 0 for both predictors on every task, because every measured value is 0. That tie is not evidence of predictive skill.

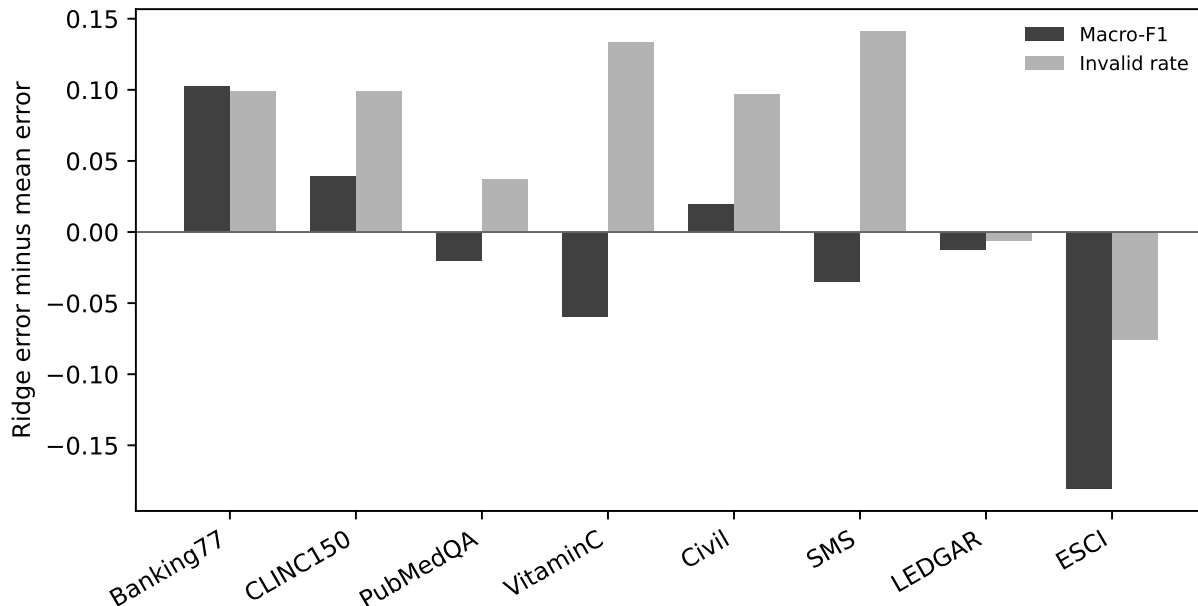


Figure 4: Ridge absolute error minus training-fold-mean absolute error, for macro-F1 and for the invalid rate. A negative bar means ridge was closer. The panel describes these eight tasks. It is not an inferential interval.

On the fit that uses all seven selector-training tasks, the largest macro-F1 coefficients are the primitive indicators: Qwen -0.385 and Laya -0.134 , against an intercept of 0.593 . The largest fingerprint coefficient on that coordinate is normalized entropy, -0.065 . On the invalid rate, the Qwen indicator is $+0.306$ and the largest fingerprint coefficient is mean label cosine, $+0.113$. The fingerprint enters as a shift applied to every primitive. Invalid outputs in this matrix occur for Qwen. A shared shift cannot raise Qwen’s invalid rate on a particular kind of task without also raising the classifier and Laya, whose invalid rates are 0.

The three largest held-out Qwen invalid rates were predicted in the wrong direction. Banking77, CLINC150 plus, and Civil Comments have rates 0.514 , 0.745 , and 1 . The unclipped ridge predictions were -0.009 , -0.073 , and -0.039 , and the unit clip made them 0. The primitive mean remained near 0.29 . The mean is closer because it retains the average Qwen failure. The fingerprint shift removes it.

The verification holdout is also a coverage fact. PubMedQA and VitaminC are the only `paired_text` tasks in the selector-training pool. Holding them out leaves the relation bit constant, so the column is zeroed. The fit cannot use a structural distinction that the training fold does not contain.

9.3 Choice

On every held-out task, and on all four policies that read a vector, the ridge profile and the primitive mean select the same primitive. That primitive is the classifier.

The measured oracle, using the same rules on the measured profile, also selects the classifier except in three cases. These cases are not successes of the predictor. They show that the objective changes the preferred primitive, and that neither predictor tracked that change.

- Highest macro-F1 on PubMedQA selects Laya: 0.3128 versus 0.2691 for the classifier.
- Highest macro-F1 on Civil Comments selects Laya: 0.7367 versus 0.6298 for the C100k classifier.
- Lowest p50 latency on SMS Spam selects the rule: 0.020 ms versus 0.188 ms for the classifier. The security fold has no rule row in training, so neither predictor can offer a rule.

Equal weights and shared weights select the classifier on every task, including when they read the measured profile. After min-max scaling on the training tasks, Laya’s latency occupies the top of the range. The gap between the rule and the classifier is small beside it, and on SMS Spam the classifier has higher macro-F1.

ESCI is the domain holdout. Ridge reduces macro-F1 error from 0.2825 to 0.1024 and invalid-rate error from 0.1199 to 0.0444. Every vector policy still selects the classifier, which is also the oracle’s selection. A closer profile did not change the decision. We do not describe this as generalization of a selector to e-commerce.

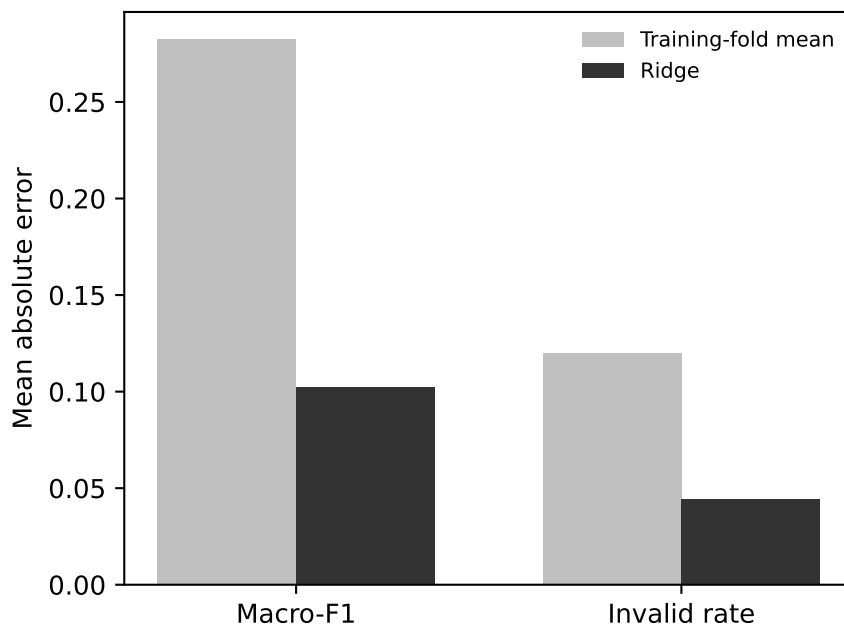


Figure 5: ESCI only. Macro-F1 error is 0.2825 for the training-fold mean and 0.1024 for ridge. Invalid-rate error is 0.1199 and 0.0444. Both predictors and the oracle select the classifier.

9.4 Generative validity

Table 3 is the main validity result. The rule, the classifier, and Laya produce no invalid outputs. Qwen’s rate depends on the task.

On Civil Comments every one of the 1,000 outputs is invalid, so macro-F1 is 0. That zero is a measured contract failure, not a hand-entered score. On Banking77 and CLINC150 plus, invalid outputs are 514 and 745 of 1,000 rows, so macro-F1 there is mostly a parse outcome. On SMS Spam, VitaminC, and ESCI, invalid outputs are 25, 23, and 32 of 1,000 rows. Macro-F1 on those tasks is mostly the legal predictions, and it is still low relative to the classifier (0.2067, 0.2209, and 0.0358).

The raw strings that failed the match were not saved. We cannot say whether they were explanations, extra words, or misspelled labels. We can say they were not exact legal labels under the pin. A later pass could store the strings. Changing the prompt, the decoder, or the parser to reduce V would be a different primitive.

10 Limitations

The task sample contains eight tasks. Intent and verification contain two tasks each. Moderation, security, and business/legal contain one. The e-commerce domain contains one held-out task. Leave-one-family-out in this sample is often leave-one-task-out. Example counts inside Banking77 or Civil Comments do not repair that.

The rule exists on one task. When that task is held out, the rule cannot be predicted. One Laya checkpoint and one Qwen checkpoint stand for their primitive families. Another decision model or another generative model could behave differently. In particular, grammar-constrained decoding can remove structural invalidity (Chavan, 2026). This study did not use it.

Latency is tied to the device that produced it. A smaller Qwen latency is not evidence that generation is cheaper than Laya in general, because Laya ran on CPU. We do not rerun the CPU primitives on GPU after seeing the matrix. Doing so would replace the frozen measurement with a second condition chosen after the outcome.

API dollars are identically zero. The study does not estimate a dollar price for local hardware and does not compare against a paid API. Training-time comparisons are dominated by the fact that only the classifier is fit on the task.

C100k is not full-data training. Civil Comments’ classifier macro-F1 of 0.6298, VitaminC’s 0.3826, and ESCI’s 0.2189 are scores of the capped fits. The fingerprint’s training size still records the full training split.

The fingerprint omits documented shift wherever the code is unspecified or blank. It has no primitive-specific slope. A shuffled-fingerprint control was discussed in an early audit note and was not part of the frozen predictor specification. The control that was run is the primitive mean. We do not add the shuffled control, or an interaction term, after seeing Table 4.

The literature screen is incomplete in the sense recorded above. A closer paper could still sit in an index we did not successfully query.

11 Threats to validity

Construct. Macro-F1, with an invalid row counted as wrong, records whether the bounded decision matched the reference label, including contract failures. V records whether the output was a legal label. Reporting only one of them would mix those questions. We report both. Accuracy is not Q . On Civil Comments the C100k classifier’s accuracy is 0.929 and its macro-F1 is 0.6298,

from `research/results/raw/performance_p0_p1.json`. That gap is why macro-F1 is the quality coordinate.

Internal. The same evaluation rows and the same input string are used for every primitive on a task. The generative device differs. Seeded subsamples for length and for C100k are pre-registered. ESCI product text is not the undownloaded official products file; the row identity is the official examples file. A reader who needs bit-identical product strings must use the documented tasksource join.

External. One generative configuration and one decision-model checkpoint do not represent their families. The label sets are closed. The study does not speak to multilabel decisions, tool use, or open-ended generation. ESCI is one e-commerce task, not a domain sample.

Statistical. The selector’s n is the number of tasks in a fold, not the number of evaluation rows. Holding out intent leaves five training tasks. Holding out a singleton leaves six. Coefficients from a fit with more fingerprint columns than tasks are descriptive of that fit. We do not interpret them as stable partial effects beyond the sign pattern already stated: primitive indicators are larger than individual fingerprint coefficients on the seven-task fit.

12 Discussion

The measured matrix shows substantial task-dependent variation among the evaluated implementations. The classifier’s macro-F1 is high on Banking77, CLINC150 plus, SMS Spam, and LEDGAR, and much lower on PubMedQA, VitaminC, and ESCI. Laya exceeds the classifier on PubMedQA and on Civil Comments under macro-F1, and is far from it on Banking77 and LEDGAR. The rule is close to the classifier on the only task where a rule was defined, and is not defined elsewhere. Qwen’s macro-F1 is the lowest eligible score on seven of the eight tasks. On VitaminC it is 0.2209 and Laya is 0.2165. The reason a large gap appears is not the same on every task. Where V is near 0, the gap is among legal labels. Where V is near 1, the gap is the contract.

The predictor results are about a model that was not allowed to represent “Qwen fails this kind of task” as a separate slope. Under that restriction, the primitive mean is a strong baseline because it already carries the level of each primitive. The fingerprint sometimes moves macro-F1 error down. It usually moves invalid-rate error up, including on the tasks where the contract failure is largest. Even the helpful ESCI movement does not change the policy. LLMRouterBench reports a related pattern inside LLM routing: a method can fail to beat a simple baseline even when model behavior is heterogeneous (Li, Zhang, et al., 2026b). We cite that as a caution, not as the same experiment.

We do not conclude that a fingerprint with primitive-specific slopes, a larger task sample, or a different generative configuration would fail. We also do not conclude that it would succeed. The protocol was frozen before Table 4 existed. Changing the model to fit the miss would spend that property.

13 Conclusion

Computational primitives exhibit substantial task-dependent differences in quality and operational behavior. In this eight-task pilot, a simple pre-deployment task fingerprint did not reliably improve prediction over primitive-specific training-fold mean baselines, and it did not alter the resulting computational choice. The invalid-output rate of the pinned generative configuration is itself a large,

task-dependent part of that behavior, from a few percent of rows to every row. Eight tasks, several of them alone in their family, do not establish that task fingerprints are unpredictable in general. They do establish what this frozen measurement found.

14 Reproducibility

Freeze identifiers are FPS-1, FPS-2, EP-1, ST-1, and SD-1, dated 26 September 2026. The fingerprint manifest, performance matrix, and selector reading are

```
research/docs/numerical_fingerprint_manifest.md
research/docs/performance_matrix.md
research/docs/selector_results.md
```

The matrix is generated from

```
research/results/raw/performance_p0_p1.json
research/results/raw/performance_dg.json
```

The selector fit and its output are

```
research/src/selector/fit_sd1.py
research/results/raw/selector_sd1.json
```

Figures are

```
research/paper/make_figures.py,
```

which reads those JSON files and does not refit the model. The SMS rule file and its hash are in Section 5. Tokenizer and encoder revisions are in Section 6. Raw corpora are local and are not redistributed with this draft. ESCI product-text provenance is stated in Section 8.

The public repository is <https://github.com/imsudip45/task-level-predictability>.

15 Author contributions

Sudip Niroula: Conceptualization, Methodology, Software, Investigation, Formal Analysis, Data Curation, Visualization, Writing – Original Draft, and Writing – Review & Editing. Mandip Pokharel: Conceptualization, Methodology, Investigation, and Writing – Review & Editing. Both authors reviewed and approved the final manuscript.

16 Conflict of interest

The authors declare no conflict of interest.

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